

Initial Project Planning

Date	15 July 2026
Team ID	SWTID-2026-7750
Project Name	Credit Card Approval Prediction System
Maximum Marks	5 Marks

Product Backlog, Sprint Schedule, and Estimation

Sprint	Functional Requirement (Epic)	User Story Number	User Story / Task	Story Points	Priority	Team Members	Sprint Start Date	Sprint End Date (Planned)
Sprint-1	Data Collection & Preprocessing	USN-1	As a developer, I want to collect the applicant and credit history datasets so that the machine learning model can be trained using reliable data.	3	High	Maddula Manasa	25-06-2026	28-06-2026
Sprint-1	Data Cleaning & Feature Engineering	USN-2	As a developer, I want to clean the dataset, handle missing values, perform feature engineering, and encode categorical features so that the data is suitable for machine learning.	5	High	Goriparthi Naga Harini	25-06-2026	30-06-2026
Sprint-2	Machine Learning Model Development	USN-3	As a developer, I want to train and compare Logistic Regression, Decision Tree, Random Forest, and XGBoost models so that the best-performing model can be selected for prediction.	8	High	Maddula Manasa	01-07-2026	05-07-2026
Sprint-2	Model Evaluation	USN-4	As a developer, I want to evaluate the trained models using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC so that the most accurate model is selected.	3	High	Nageswari Kondru	01-07-2026	05-07-2026

Sprint-3	Web Application Development	USN-5	As a user, I want to enter my personal and financial details through a Flask web application and instantly receive a credit card approval prediction with a confidence score.	8	High	Nageswari Kondru, Maddula Manasa, Goriparthi Naga Harini	06-07-2026	10-07-2026
Sprint-3	Model Integration & Deployment	USN-6	As a developer, I want to integrate the trained Random Forest model with the Flask application, test the system, deploy it on Render, and upload the project to GitHub so that users can access the application online.	5	High	Nageswari Kondru	08-07-2026	10-07-2026